Forward and Backward Pass of a Log-Barrier ADMM QP Solver

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1 From trajectory optimization to a QP

We consider the discrete-time optimal control problem (OCP) in the form of [1],

$$\begin{aligned}\min_{(x_t,u_t)_{t=0}^N} \quad & \textstyle\sum_{t=0}^{N-1} \ell_t(x_t,u_t) + \ell_N(x_N) \\ \text{s.t.} \quad & x_0 = x_{\text{init}}, \\ & f_t(x_t,u_t,x_{t+1},u_{t+1}) = 0, \quad t = 0,\dots,N-1, \\ & \underline{g}_t \le g_t(x_t,u_t) \le \overline{g}_t, \qquad t = 0,\dots,N,\end{aligned} \tag{1}$$

with states $x_t \in \mathbb{R}^{n_x}$, controls $u_t \in \mathbb{R}^{n_u}$, implicit discretized dynamics f_t, and pointwise-in-time inequality constraints g_t (bounds, obstacles, friction cones, ...). The OCP data depend on parameters θ (cost weights, constraint geometry) with respect to which we ultimately want gradients.

NLP. Overloading $x := (x_t,u_t)_{t=0}^N \in \mathbb{R}^{n_w}$ for the stacked stage variables, collect the cost into ℓ, the initial-condition and dynamics residuals into $f \in \mathbb{R}^m$, and the inequalities into *one-sided* rows $g \in \mathbb{R}^p$ (each finite bound of (1) contributes one row, $g_t \le \overline{g}_t$ or $-g_t \le -\underline{g}_t$; infinite bounds are dropped), so that (1) is the parametric nonlinear program

$$\min_x \ \ell(x;\theta) \quad \text{s.t.} \quad f(x;\theta) = 0, \quad g(x;\theta) \le h(\theta). \tag{2}$$

SQP. Sequential quadratic programming [8, Ch. 18] solves (2) by iterating: at the current primal–dual iterate, quadratize the Lagrangian, linearize the constraints, solve the resulting QP, and update the iterate along its solution with a linesearch. With the constraint Jacobians $C := \partial f/\partial x$ and $G := \partial g/\partial x$ evaluated at the iterate, $P \approx \nabla^2_{xx}\mathcal{L}$ a Hessian approximation (in practice Gauss–Newton), and q the corresponding gradient, each subproblem is the convex QP studied in the rest of this note:

$$\text{QP}(\theta): \quad \min_x \ \tfrac{1}{2}x^\top P x + q^\top x \quad \text{s.t.} \quad Cx = c, \quad Gx \le h, \tag{3}$$

with $P \succeq 0$, $C \in \mathbb{R}^{m\times n_w}$, $G \in \mathbb{R}^{p\times n_w}$. The time structure of (1) makes P block-tridiagonal, C block-bidiagonal, and G block-diagonal [1] — structure both passes below exploit.

2 Perturbed KKT and smoothed complementarity

With multipliers $y_f \in \mathbb{R}^m$ (equalities) and $y_g \in \mathbb{R}^p$ (inequalities) and the slack $s := h - Gx$, the KKT conditions of (3) are

$$\begin{aligned} & Px + q + C^\top y_f + G^\top y_g = 0, \qquad Cx = c, \\ & Gx + s = h, \qquad s \ge 0, \quad y_g \ge 0, \quad s \circ y_g = 0, \end{aligned} \tag{4}$$

where $\circ$ is the elementwise product. All difficulty — combinatorial for the solver, and geometric for differentiation — sits in the last line. The solution map $\theta \mapsto (x, y_f, y_g)$ of (4) is only piecewise smooth: at points where *strict complementarity* fails ($s_i = y_{g,i} = 0$ for some row) the KKT Jacobian is singular and the map is nondifferentiable, with gradient jumps across active-set changes [6, 7]. We therefore smooth (4) with two perturbations, used together or separately.

(i) Barrier perturbation κ. Following the interior-point tradition, replace complementarity by the *perturbed KKT* (central-path) condition [8, Ch. 19]

$$s \circ y_g = \kappa \mathbf{1}, \qquad s > 0, \ y_g > 0, \tag{5}$$

for a barrier parameter $\kappa > 0$. This is the stationarity system of the equality-constrained log-barrier subproblem (add $-\kappa \sum_i \log s_i$ to the cost); as $\kappa \searrow 0$ it recovers (4). Differentiating the κ-perturbed system instead of (4) is exactly the mechanism used to obtain smooth gradients through QP/NLP layers [5, 7].

(ii) Inequality-slack (Moreau) perturbation (δ_ξ, γ_ξ). Following TurboMPC's slack convention [1], introduce an *inequality slack* $\xi \ge 0$ per one-sided row, toggled by the binary indicator $\delta_\xi \in \{0, 1\}$ and penalized quadratically: $Gx - \delta_\xi \xi \le h$ with penalty $\delta_\xi \frac{\gamma_\xi}{2}\|\xi\|^2$, $\gamma_\xi > 0$. For $\delta_\xi = 1$, minimizing over ξ first replaces the indicator of $\{Gx \le h\}$ by its Moreau envelope $\frac{\gamma_\xi}{2}\operatorname{dist}^2(Gx, \{\cdot \le h\})$ [4]: the constraint becomes a stiff quadratic penalty, the QP stays feasible under SQP linearization error, and the multiplier increase continuously as $y_g = \gamma_\xi \xi$ (TurboMPC's `ADMMSlack` smoothing [1]). I'm not sure this is true since we are potentially using an implicit integrator which can lead to infeasible/conflicting equality constraints.

Combined perturbed system. Applying both, the log-barrier subproblem is

$$\begin{aligned} \min_{x,\xi}\ & \tfrac{1}{2}x^\top Px + q^\top x + \delta_\xi \frac{\gamma_\xi}{2}\|\xi\|^2 - \kappa \sum_{i=1}^{p} \log s_i \\ \text{s.t. } & Cx = c, \qquad s = h - Gx + \delta_\xi\, \xi, \end{aligned} \tag{6}$$

whose stationarity in s gives $y_g = \kappa/s$ (elementwise) and, when $\delta_\xi = 1$, stationarity in ξ gives $\gamma_\xi \xi = y_g$. Eliminating $\xi = \gamma_\xi^{-1} y_g$, the optimality conditions collapse to a square smooth root-finding problem in $w := (x, y_f, y_g)$:

$$F_{\mathrm{QP}}(w;\theta) := \begin{bmatrix} Px + q + C^\top y_f + G^\top y_g \\ Cx - c \\ s \circ y_g - \kappa \mathbf{1} \end{bmatrix} = 0, \tag{7}$$

$$s := h - Gx + \delta_\xi \gamma_\xi^{-1} y_g \ (>0), \qquad y_g > 0. \tag{8}$$

The forward pass converges to the solution of (7) for each QP subproblem; the backward pass differentiates the nonlinear counterpart of this system (Section 4). The two perturbations place the solver in a four-way family of ADMM methods:

	$\kappa = 0$ (hard compl.)	$\kappa > 0$ (central path)
$\delta_\xi = 0$	standard KKT (4); OSQP [2] (why OSQP?)	pure barrier
$\delta_\xi = 1$	Moreau; `ADMMSlack` [1]	slack barrier

For any $\kappa > 0$ the residual F_{QP} is smooth and its Jacobian is nonsingular (Section 4), so $\theta \mapsto w(\theta)$ is C^1 — including across activation/deactivation of constraints. The price is an $O(\kappa)$ bias of $x(\theta)$ relative to the hard QP solution, and an $O(\|y_g\|/\gamma_\xi)$ constraint violation from the inequality slack when $\delta_\xi = 1$; $\kappa \to 0$ (with $\delta_\xi = 0$, or $\gamma_\xi \to \infty$) recovers (4).

3 Forward pass: ADMM with a log-barrier slack update

We follow the ADMM convention of [1]. Stack the constraints as $A := \left[\begin{smallmatrix} C \\ G \end{smallmatrix}\right]$ and give Ax its own copy $z = (z_f, z_g)$:

$$\begin{aligned} \min_{x,z,\xi}\ & \tfrac{1}{2}x^\top Px + q^\top x + \delta_\xi \frac{\gamma_\xi}{2}\|\xi\|^2 + \mathcal{I}_{\{c\}}(z_f) + B_\kappa(z_g, \xi) \\ \text{s.t. } & Ax - z = 0, \end{aligned} \tag{9}$$

where $\mathcal{I}$ denotes an indicator function and

$$B_\kappa(z_g, \xi) := -\kappa \sum_{i=1}^{p} \log\big(h_i - z_{g,i} + \delta_\xi\, \xi_i\big) \tag{10}$$

is the log barrier on the copy. (At $\kappa = 0$, B_κ degenerates to the indicator of $\{z_g - \delta_\xi \xi \le h\}$ and (9) is exactly the TurboMPC splitting.) Introducing duplicated variables $(\tilde{x}, \tilde{z})$ as in [1], multipliers w for $\tilde{x} - x = 0$ (which vanishes identically after the first iteration and is dropped below) and $y = (y_f, y_g)$ for $\tilde{z} - z = 0$, and step-size parameters $\sigma > 0$, $\rho := \mathrm{blkdiag}(\rho_f I_m,\ \rho_g I_p) \succ 0$, the augmented Lagrangian is

$$\begin{aligned} \mathcal{L}_{\sigma,\rho} = {} & \tfrac{1}{2}\tilde{x}^\top P\tilde{x} + q^\top \tilde{x} + \delta_\xi \frac{\gamma_\xi}{2}\|\xi\|^2 + \mathcal{I}_{\{c\}}(z_f) + B_\kappa(z_g, \xi) \\ & + \mathcal{I}_{\{0\}}(A\tilde{x} - \tilde{z}) + \frac{\sigma}{2}\big\|\tilde{x} - x\big\|^2 + \tfrac{1}{2}\big\|\tilde{z} - z + \rho^{-1} y\big\|_\rho^2, \end{aligned} \tag{11}$$

with $\|v\|_\rho^2 := v^\top \rho\, v$. One ADMM iteration minimizes $\mathcal{L}_{\sigma,\rho}$ over $(\tilde{x}, \tilde{z})$, then over (x, z, ξ), then ascends the duals, with over-relaxation $\alpha \in (0, 2)$.

3.1 Primal update

Minimizing (11) over $(\tilde{x}, \tilde{z})$ is an equality-constrained strictly convex QP whose KKT conditions are the linear system [1, Eq. (7)]

$$\begin{bmatrix} P + \sigma I & A^\top \\ A & -\rho^{-1} \end{bmatrix} \begin{bmatrix} \tilde{x}^{k+1} \\ \nu^{k+1} \end{bmatrix} = \begin{bmatrix} \sigma x^k - q \\ z^k - \rho^{-1} y^k \end{bmatrix}. \tag{12}$$

The $-\rho^{-1}$ block is eliminated by a Schur complement:

$$S\,\tilde{x}^{k+1} = \sigma x^k - q + A^\top\big(\rho z^k - y^k\big), \qquad \tilde{z}^{k+1} = A\tilde{x}^{k+1}, \tag{13}$$

with $S := P + \sigma I + A^\top \rho A \succ 0$. By the structure noted in Section 1, S is block-tridiagonal in time; its sparse factorization is computed once and reused across iterations, refactorizing only when ρ is adapted [1, 2]. The over-relaxed iterates are

$$\hat{x}^{k+1} = \alpha \tilde{x}^{k+1} + (1-\alpha) x^k, \quad \hat{z}^{k+1} = \alpha \tilde{z}^{k+1} + (1-\alpha) z^k, \tag{14}$$

and the x-minimization gives $x^{k+1} = \hat{x}^{k+1}$.

3.2 Slack update: the proximal operator of the log barrier

Minimizing (11) over z_f gives $z_f^{k+1} = c$ (so z_f need not be stored). The inequality block is the essential step. Let

$$\bar{z}_g^{k+1} := \hat{z}_g^{k+1} + \rho_g^{-1} y_g^k \tag{15}$$

be the usual ADMM target — the same point OSQP projects onto $\{z_g \le h\}$ and TurboMPC's `ADMMSlack` projects smoothly. Here the update is the prox of the barrier, jointly over (z_g, ξ):

$$(z_g, \xi)^{k+1} = \arg\min_{z,\xi}\ B_\kappa(z, \xi) + \delta_\xi \tfrac{\gamma_\xi}{2}\|\xi\|^2 + \tfrac{\rho_g}{2}\|z - \bar{z}_g^{k+1}\|^2 \tag{16}$$

(for $\delta_\xi = 0$ the inequality slack is fixed, $\xi \equiv 0$, exactly as in TurboMPC's slack update). Per row, with the barrier slack $s(z, \xi) := h - z + \delta_\xi \xi$, the objective of (16) is

$$\varphi(z, \xi) = -\kappa \log s(z, \xi) + \delta_\xi \tfrac{\gamma_\xi}{2}\xi^2 + \tfrac{\rho_g}{2}\big(z - \bar{z}_g^{k+1}\big)^2. \tag{17}$$

With $\partial s/\partial z = -1$ and $\partial s/\partial \xi = \delta_\xi$, the chain rule gives

$$\begin{aligned} \partial_z \varphi &= \tfrac{\kappa}{s} + \rho_g\big(z - \bar{z}_g^{k+1}\big) = 0 \quad &\Rightarrow\ \tfrac{\kappa}{s} = \rho_g\big(\bar{z}_g^{k+1} - z\big), \\ \partial_\xi \varphi &= -\tfrac{\kappa}{s} + \gamma_\xi \xi = 0 \ \ (\delta_\xi{=}1) \quad &\Rightarrow\ \gamma_\xi \xi = \tfrac{\kappa}{s}, \end{aligned} \tag{18}$$

and the stationary $\xi = \kappa/(\gamma_\xi s) > 0$ is automatically positive. Only true for $s > 0$, which is not shown at this point yet. Solving (18) for $z = \bar{z}_g^{k+1} - \kappa/(\rho_g s)$ and $\delta_\xi \xi = \delta_\xi \kappa/(\gamma_\xi s)$ and substituting back into $s = h - z + \delta_\xi \xi$ with the residual $r^{k+1} := h - \bar{z}_g^{k+1}$ yields $s = r^{k+1} + \Gamma/s$, i.e. the scalar quadratic $s^2 - r^{k+1}s - \Gamma = 0$ per row, whose positive root is, elementwise,

$$\boxed{\begin{aligned} s^{k+1} &= b_\Gamma\big(r^{k+1}\big), \qquad & \Gamma := \kappa\big(\rho_g^{-1} + \delta_\xi\, \gamma_\xi^{-1}\big), \\ \xi^{k+1} &= \delta_\xi\, \frac{\kappa}{\gamma_\xi\, s^{k+1}}, \quad & z_g^{k+1} = h - s^{k+1} + \xi^{k+1}, \end{aligned}} \tag{19}$$

where b_μ is the *retraction map* (positive root of $b^2 - vb - \mu = 0$)

$$b_\mu(v) := \frac{v + \sqrt{v^2 + 4\mu}}{2}, \qquad \mu > 0, \tag{20}$$

with the elementary properties:

(i) $b_\mu(v) > 0$ for all $v \in \mathbb{R}$;
(ii) $b_\mu(v)\, b_\mu(-v) = \mu$ (built-in complementarity);
(iii) $b_\mu(v) - b_\mu(-v) = v$;
(iv) $b_\mu(v) \to \max(v, 0)$ as $\mu \to 0^+$ (smooth ReLU);
(v) $b'_\mu(v) = \frac{1}{2}\big(1 + v/\sqrt{v^2+4\mu}\big) \in (0,1)$ and $b'_\mu(v) + b'_\mu(-v) = 1$.

In the default mode $\delta_\xi = 0$: $\Gamma = \kappa/\rho_g$, $\xi^{k+1} = 0$, and $z_g^{k+1} = h - b_{\kappa/\rho_g}(r^{k+1})$. (For $v < 0$ we evaluate $b_\mu(v) = 2\mu/(\sqrt{v^2 + 4\mu} - v)$ to avoid numerical issues.)

3.3 Dual update and exact perturbed complementarity

The dual ascent uses the over-relaxed copy [1]:

$$y_f^{k+1} = y_f^k + \rho_f\big(\hat{z}_f^{k+1} - c\big), \quad y_g^{k+1} = y_g^k + \rho_g\big(\hat{z}_g^{k+1} - z_g^{k+1}\big). \tag{21}$$

By (15), $y_g^{k+1} = \rho_g(\bar{z}_g^{k+1} - z_g^{k+1})$, and the z-stationarity in (18) evaluates this in closed form:

$$\boxed{y_g^{k+1} = \frac{\kappa}{s^{k+1}} \ \Rightarrow\ s^{k+1} \circ y_g^{k+1} = \kappa \mathbf{1},\ \ \gamma_\xi \xi^{k+1} = \delta_\xi y_g^{k+1}} \tag{22}$$

elementwise, *at every iteration* k — not only in the limit. The iterates live on the central-path manifold (5) by construction, with $y_g^{k+1} > 0$ automatic from property (i), and the inequality-slack stationarity holds identically in both modes. At a fixed point, $Ax = z$ makes $s = h - Gx + \delta_\xi \xi = h - Gx + \delta_\xi \gamma_\xi^{-1} y_g$, so (x, y_f, y_g) satisfies *exactly* the perturbed KKT system (7)–(8) of the QP.

Limits. As $\kappa \to 0^+$, property (iv) collapses (19) to TurboMPC's smoothed projection (for $\delta_\xi = 1$), $\tilde{\Pi}^{\gamma_\xi}(\tilde{z}_g) = \frac{\rho_g}{\gamma_\xi + \rho_g}\tilde{z}_g + \frac{\gamma_\xi}{\gamma_\xi + \rho_g}\Pi(\tilde{z}_g)$ [1, Eq. (15)], and, for $\delta_\xi = 0$, to OSQP's hard projection $z_g^{k+1} = \min(\tilde{z}_g^{k+1}, h)$: a smooth deformation of the standard projection ADMM, at identical per-iteration cost.

Termination, ρ-adaptation, continuation. Convergence is declared on the standard residuals $r_{\text{prim}} = \|Ax - z\|_\infty$ and $r_{\text{dual}} = \|Px + q + A^\top y\|_\infty$ [1, Eq. (3)] (slack stationarity $\gamma_\xi \xi - \delta_\xi y_g = 0$ holds identically by (22)); $\bar{\rho}$ is adapted OSQP-style from the residual ratio with $\rho_f = 10^3 \bar{\rho}$, $\rho_g = \bar{\rho}$, $\sigma = 10^{-6}$, refactorizing S on each change. TODO: 1. Central path iterations for κ.
2. Check if it's better to use fixed κ for differentiable MPC.
3. Check if the residuals should be defined differently.

4 Backward pass: implicit differentiation of the NLP's perturbed KKT

Let $L = L(x^\star)$ be a scalar loss of the converged SQP solution. Here, the object that carries the sensitivities of (2) is the *NLP's* own smoothed KKT system, and QP-level sensitivities coincide with the NLP's only when the QP carries the *exact* Lagrangian Hessian at the solution [7, Thm. 2]. Since the forward SQP typically runs a Gauss–Newton P (Section 1), differentiating the QP it actually solved can degrade the gradients [7, Rem. 3]. The backward pass therefore re-assembles the NLP system at the converged point.

4.1 The NLP's smoothed KKT system

At SQP convergence, the iterate $w^\star := (x^\star, y_f, y_g)$ satisfies the $(\kappa, \delta_\xi, \gamma_\xi)$-perturbed optimality conditions of (2) — the nonlinear counterpart of (7), with the smoothed complementarity imposed on the *nonlinear* residual (the interior-point-smoothed conditions of [7, Eq. (10)]:

$$F(w;\theta) := \begin{bmatrix} \nabla_x \ell(x;\theta) + C(x)^\top y_f + G(x)^\top y_g \\ f(x;\theta) \\ s \circ y_g - \kappa \mathbf{1} \end{bmatrix} = 0, \tag{23}$$

$$s := h(\theta) - g(x;\theta) + \delta_\xi \gamma_\xi^{-1} y_g \ (> 0), \qquad y_g > 0, \tag{24}$$

where $C(x) := \partial f/\partial x$ and $G(x) := \partial g/\partial x$ are the constraint Jacobians evaluated at x (for affine f, g and quadratic ℓ, (23) reduces exactly to (7)). The first row is ∇_x of the NLP Lagrangian $\mathcal{L} := \ell(x;\theta) + y_f^\top f(x;\theta) + y_g^\top g(x;\theta)$. Differentiating (23) at its root yields the gradient of the relaxed solution map, smooth across active-set changes.

4.2 The implicit function theorem

The residual $F(w;\theta)$ of (23) is C^1 in (w,θ) whenever ℓ, f, g are twice continuously differentiable [7, Ass. 1]. Let $w^\star$ satisfy $F(w^\star;\theta)=0$ with $s>0$, $y_g>0$ (guaranteed for $\kappa>0$ by (22); at SQP convergence the linearized and nonlinear residuals coincide), and define $M := \partial F/\partial w\big|_{(w^\star,\theta)}$. If M is nonsingular, the implicit function theorem yields a locally unique C^1 map $\theta \mapsto w(\theta)$ with $F(w(\theta);\theta)\equiv 0$ and $\frac{\partial F}{\partial w}\frac{\partial w}{\partial \theta}+\frac{\partial F}{\partial \theta}=0$,

$$\frac{\partial w}{\partial \theta} = -\,M^{-1}\frac{\partial F}{\partial \theta} \tag{25}$$

— the sensitivity result of [7, Thm. 1]; their Thm. 3 adds that the smoothed map is C^1 for small κ and approaches the unsmoothed one with error $O(\kappa)$. The gradient of a scalar loss needs a *single* solve with $M^\top$: absorbing the sign, define the adjoint $\lambda=(\lambda_x,\lambda_f,\lambda_g)$ by

$$M^\top\lambda = -\big(\nabla_x L,\,0,\,0\big), \qquad \nabla_\theta L = \Big(\frac{\partial F}{\partial \theta}\Big)^{\!\top}\lambda. \tag{26}$$

4.3 The Jacobian M and the adjoint system

Differentiating (23) with $\partial s/\partial x = -G(x)$ and $\partial s/\partial y_g = \delta_\xi\gamma_\xi^{-1}I$ gives, in block form (rows = residuals, columns = x, y_f, y_g; all quantities evaluated at $w^\star$),

$$M = \begin{bmatrix} H & C^\top & G^\top \\ C & 0 & 0 \\ -\operatorname{diag}(y_g)\,G & 0 & \operatorname{diag}\big(s+\delta_\xi\gamma_\xi^{-1}y_g\big)\end{bmatrix}, \tag{27}$$

where the $(1,1)$ block is the *exact Lagrangian Hessian* of the NLP,

$$H := \nabla^2_{xx}\mathcal{L} = \nabla^2_{xx}\ell + \sum_{j=1}^{m} y_{f,j}\,\nabla^2_{xx}f_j + \sum_{i=1}^{p} y_{g,i}\,\nabla^2_{xx}g_i, \tag{28}$$

Because f and g are stagewise, H retains P's block-tridiagonal sparsity. The adjoint system (26) written out is

$$H\lambda_x + C^\top\lambda_f - G^\top\operatorname{diag}(y_g)\,\lambda_g = -\nabla_x L, \tag{29a}$$
$$C\lambda_x = 0, \tag{29b}$$
$$G\lambda_x + \operatorname{diag}\big(s+\delta_\xi\gamma_\xi^{-1}y_g\big)\lambda_g = 0. \tag{29c}$$

Two equivalent reductions of (29) can be used in practice.

4.4 Reduction A: retraction parameterization

On the central-path manifold (stated for the default $\delta_\xi=0$), properties (ii)–(iii) of b_κ make the pair (s,y_g) the graph of a *single* unconstrained variable $v := y_g - s \in \mathbb{R}^p$:

$$s = b_\kappa(-v), \qquad y_g = b_\kappa(v), \tag{30}$$

which satisfies $s\circ y_g = \kappa\mathbf{1}$, $s, y_g>0$ *identically* — the complementarity residual disappears from the system. Differentiating with $B_p := \operatorname{diag}\big(b'_\kappa(v)\big)$, $B_n := \operatorname{diag}\big(b'_\kappa(-v)\big)$ (both positive by property (v)) gives $dy_g = B_p\,dv$, $ds = -B_n\,dv$, and the remaining residuals (stationarity, primal inequality $g(x;\theta)+s-h$, equality) linearize in (dx, dv, dy_f) with matrix $\left[\begin{smallmatrix} H & G^\top B_p & C^\top \\ G & -B_n & 0 \\ C & 0 & 0\end{smallmatrix}\right]$. Subtracting $G^\top\times$(second row) from the first and using $B_p+B_n=I$ (property (v)) symmetrizes it:

$$J = \begin{bmatrix} H-G^\top G & G^\top & C^\top \\ G & -B_n & 0 \\ C & 0 & 0\end{bmatrix} = J^\top. \tag{31}$$

J is symmetric, and the adjoint solve is

$$J\begin{bmatrix}\lambda_x\\ \lambda_v\\ \lambda_f\end{bmatrix} = -\begin{bmatrix}\nabla_x L\\ 0\\ 0\end{bmatrix}, \qquad \bar\lambda_g := B_p\,\lambda_v, \tag{32}$$

where $\bar\lambda_g$ is the effective inequality adjoint used in the gradient assembly below.

4.5 Reduction B: Schur elimination onto a weighted Hessian

Alternatively, first rewrite the inequality-adjoint block in the standard interior-point saddle form. Let

$$D := \operatorname{diag}\big(s+\delta_\xi\gamma_\xi^{-1}y_g\big), \quad Y := \operatorname{diag}(y_g), \quad W := DY^{-1}.$$

Introducing the effective inequality adjoint

$$\bar\lambda_g := -Y\lambda_g, \tag{33}$$

the adjoint equations (29a)–(29c) can be written as the lifted symmetric saddle system

$$\begin{bmatrix} H & C^\top & G^\top \\ C & 0 & 0 \\ G & 0 & -W\end{bmatrix}\begin{bmatrix}\lambda_x\\ \lambda_f\\ \bar\lambda_g\end{bmatrix} = -\begin{bmatrix}\nabla_x L\\ 0\\ 0\end{bmatrix}. \tag{34}$$

Thus the inequality Jacobian G appears unscaled in the off-diagonal blocks, while the slack–multiplier ratio appears in the lower-right block. Eliminating $\bar\lambda_g$ from (34) gives

$$\bar\lambda_g = W^{-1}G\lambda_x.$$

Equivalently, eliminating λ_g directly from (29c), whose coefficient D is diagonal and positive for $\kappa>0$, gives

$$\lambda_g = -D^{-1}G\,\lambda_x. \tag{35}$$

Substituting into (29a) folds the inequalities into the Hessian block:

$$\begin{bmatrix} H+G^\top W^{-1}G & C^\top \\ C & 0\end{bmatrix}\begin{bmatrix}\lambda_x\\ \lambda_f\end{bmatrix} = -\begin{bmatrix}\nabla_x L\\ 0\end{bmatrix}. \tag{36}$$

Equivalently, (36) is the Schur complement of the lower-right block $-W$ in (34). This reduced saddle system is exactly the symmetric reduced coefficient matrix of [7, Eq. (9)] (there $S^{-1}M = \operatorname{diag}(\mu/s)$ plays the role of W^{-1}). W^{-1} is the C^1 analogue of a hard active-set mask:

regime	$y_{g,i}$	s_i	W_i^{-1}
inactive row	$\to 0$	> 0	$\to 0$ (row drops out)
active, $\kappa{\to}0$, $\delta_\xi{=}1$	> 0	$\to \gamma_\xi^{-1} y_{g,i}$	$\to \gamma_\xi$ (penalty stiffness)
active, $\kappa{\to}0$, $\delta_\xi{=}0$	> 0	$\to 0$	$\to \infty$ (hard constraint)

so $W^{-1} \in [0, \gamma_\xi]$ when $\delta_\xi = 1$. Since G is block-diagonal per stage, $G^\top W^{-1} G$ perturbs only the diagonal blocks of H: (36) has the same block-tridiagonal saddle structure as the forward Schur system (13).

Why do we need the smoothing κ : (36) is nonsingular iff C has full row rank (LICQ; TODO: check if this is always true for dynamics constraints) and $H + G^\top \operatorname{diag}(W) G \succ 0$ on $\ker C$, which for small κ follows from the second-order sufficient conditions at the solution [7, Ass. 1, Thm. 1]. In the hard limit $\kappa = 0$ a failure is *weakly active* row, $s_i = 0$ *and* $y_{g,i} = 0$ (strict complementarity fails): the i-th complementarity row of M vanishes identically, M is singular, and the solution map is nondifferentiable [6, 7]. The barrier removes this failure — $s \circ y_g = \kappa$ makes W smooth in y_g — so $\nabla_\theta L$ is continuous across active-set changes [5].

4.6 Assembling the parameter gradients

With (λ_x, λ_f) from (36) (or (32)) and $\bar\lambda_g$ from (33), the contraction $\nabla_\theta L = (\partial F/\partial\theta)^\top \lambda$ of (26) takes, at the NLP level, the compact form

$$\nabla_\theta L = \big[\partial_\theta \nabla_x \mathcal{L}\big]^\top \lambda_x + \big[\partial_\theta f\big]^\top \lambda_f + \big[\partial_\theta (g-h)\big]^\top \bar\lambda_g, \quad (37)$$

three mixed vector–Jacobian products of the problem functions evaluated at $(w^\star, \theta)$ — the adjoint sensitivity of [7, Eq. (13)]. For parameters that enter only the cost, $\partial_\theta f = 0$ and $\partial_\theta(g - h) = 0$, and (37) collapses to the single contraction $\nabla_\theta L = [\partial_\theta \nabla_x \ell(x^\star;\theta)]^\top \lambda_x$, one vector–Jacobian product of the cost gradient.

QP-layer special case. When the layer is the QP (3) itself, with data $\theta = (P, q, C, c, G, h)$ and $H = P$, expanding (37) datum-by-datum gives the familiar table

$$\begin{aligned} \nabla_P L &= \tfrac{1}{2}\big(\lambda_x x^{\star\top} + x^\star \lambda_x^\top\big), & \nabla_q L &= \lambda_x, \\ \nabla_C L &= \lambda_f x^{\star\top} + y_f \lambda_x^\top, & \nabla_c L &= -\lambda_f, \\ \nabla_G L &= \bar\lambda_g x^{\star\top} + y_g \lambda_x^\top, & \nabla_h L &= -\bar\lambda_g. \end{aligned} \quad (38)$$

4.7 Regularized sensitivities

The lifted system (34) also suggests a regularized sensitivity operator. Let $d_x, d_f, \bar d_g$ denote the primal, equality-multiplier, and effective inequality-multiplier sensitivities with respect to a perturbation $d\theta$. The unregularized lifted linearization has the form

$$\begin{bmatrix} H & C^\top & G^\top \\ C & 0 & 0 \\ G & 0 & -W \end{bmatrix} \begin{bmatrix} d_x \\ d_f \\ \bar d_g \end{bmatrix} = - \begin{bmatrix} r_x \\ r_f \\ r_g \end{bmatrix}, \quad (39)$$

where r_x, r_f, r_g collect the corresponding θ-derivatives of the KKT residual.

We regularize the primal and equality blocks by replacing (39) with

$$\begin{bmatrix} H + \sigma_x I & C^\top & G^\top \\ C & -\sigma_f I & 0 \\ G & 0 & -W \end{bmatrix} \begin{bmatrix} d_x \\ d_f \\ \bar d_g \end{bmatrix} = - \begin{bmatrix} r_x \\ r_f \\ r_g \end{bmatrix}, \quad (40)$$

with $\sigma_x, \sigma_f \ge 0$. The inequality block is left unchanged.

Eliminating the dual sensitivities gives the primal sensitivity as the solution of the regularized quadratic problem

$$\begin{aligned} \min_{d_x} \quad & \frac{1}{2} d_x^\top H d_x + r_x^\top d_x + \frac{1}{2}\|G d_x + r_g\|^2_{W^{-1}} \\ & + \frac{\sigma_x}{2}\|d_x\|^2 + \frac{1}{2\sigma_f}\|C d_x + r_f\|^2. \end{aligned} \quad (41)$$

Here $\|v\|_A^2 := v^\top A v$, and the convention is that $\sigma_f = 0$ recovers the hard linearized equality constraint

$$C d_x + r_f = 0. \quad (42)$$

This gives the desired interpretation. The term $\frac{\sigma_x}{2}\|d_x\|^2$ penalizes the size of the primal sensitivity, whereas $\frac{1}{2\sigma_f}\|C d_x + r_f\|^2$ penalizes violation of the linearized equality constraint. Thus smaller σ_f enforces tangent feasibility more strongly, and larger σ_f allows a softer, more stable sensitivity. The inequality contribution remains the interior-point weighted penalty $\frac{1}{2}\|G d_x + r_g\|^2_{W^{-1}}$.

The equality-multiplier sensitivity d_f also acquires a useful interpretation in the regularized system. The second block row of (40) is

$$C d_x + r_f = \sigma_f d_f. \quad (43)$$

Thus $\sigma_f d_f$ is exactly the residual in the linearized equality constraint. In the unregularized limit $\sigma_f \to 0$, this residual is forced to vanish and we recover the hard tangent condition $C d_x + r_f = 0$. Hence, d_f remains the equality-multiplier sensitivity in the stationarity equation, but also acts as a scaled slack variable measuring linearized equality infeasibility.

The regularized adjoint used for gradient computation is obtained by applying the transpose of the same regularized operator:

$$\begin{bmatrix} H + \sigma_x I & C^\top & G^\top \\ C & -\sigma_f I & 0 \\ G & 0 & -W \end{bmatrix}^\top \begin{bmatrix} \lambda_x \\ \lambda_f \\ \bar\lambda_g \end{bmatrix} = - \begin{bmatrix} \nabla_x L \\ 0 \\ 0 \end{bmatrix}. \quad (44)$$

Since the lifted operator is symmetric, this has the same matrix as (40). The resulting gradient is therefore the adjoint of the regularized sensitivity map defined by (41).

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